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Internationalizing Doctoral Education: Examining the Relationship Between International Engagement Opportunities and Research Production

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Abstract International research collaboration and international mobility are widely recognized as effective ways to improve the quantity and quality of scholarly outputs, yet it is not clear how doctoral students contribute to overall research production through international collaboration or how these opportunities may help doctoral students develop their own research publications. This article examined the relationship between doctoral students' publication outputs and the opportunities for international research collaborations and study abroad. We analyzed data from the Canadian Graduate and Professional Survey (CGPSS) and focused on two key types of international opportunities: the opportunity for international research collaboration, and the opportunity to study abroad. Our findings reveal a significantly positive relationship between the opportunity of international research collaboration and doctoral students' publication outputs. In contrast, the effects of study abroad are more complex and less pronounced. Additionally, we found that international students and students majoring in STEM disciplines tend to benefit less from these opportunities compared to their peers. We offer recommendations for future research and implications for students, institutions, and nations to improve doctoral education and research production.

Keywords Graduate education · International research collaboration · Study abroad · Publication · Research policy

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Introduction

During the first half of the 20th century knowledge production and scientific communication proliferated exponentially, and the world was said to have transitioned from an era of "little science" to an era of "big science" (de Solla Price 1961, 1963). However, exponential growth is not easy to sustain, and de Solla Price (1963) doubted that science would continue to reach new heights in the latter half of the 20th century. Contrary to his prediction, the world achieved what some sociologists have termed "global-mega science" (Baker and Powell 2024; Powell et al. 2017). Global-mega science was possible through the spread of the university-science model, where university-based researchers unlocked new discoveries by collaborating with networks of co-authors around the globe. For instance, the global share of scientific papers with international co-authors doubled during the 1990s and 2000s (Wagner et al. 2015), which network analyses revealed included expansion at the country-, researcher-, and academic subject-level (Chen et al. 2019; Ribeiro et al. 2018; Wagner et al. 2017).

The era of global-mega science coincided with the founding of new universities, the creation of new doctoral programs, and expanding enrollments within existing doctoral programs (Fernandez and Baker 2017; Fernandez et al. 2021). Therefore, since the latter half of the 20th century, doctoral training has occurred in a landscape marked by increasing science production and international research collaboration (Powell et al. 2017). While research suggested that doctoral education was an important factor that facilitated the emergence of unprecedented science production in the 21st century, there has been relatively little research that examines whether doctoral students are given opportunities to work internationally and whether having opportunities to work internationally influences doctoral students' publishing. If internationalization is one of the factors underpinning mega-global science, then it is important to understand how opportunities for international collaboration may contribute to or enhance doctoral training of future faculty and independent researchers.

We use data from the 2019 Canadian Graduate and Professional Student Survey (CGPSS) to explore how opportunities for international research collaboration (IRC) and opportunities to study abroad relate to doctoral students' publication output. We focus on Canada as one of the key contributors to global scientific research production through the 20th and early 21st centuries (Powell et al. 2017). CGPSS is an extensive dataset that surveys graduate students about various facets of their training. We estimate binary, ordinal, and multinomial logit regression models to explore how international opportunities influence the likelihood of students publishing as first authors, as well as the quantity of their publications.

By analyzing patterns of collaboration and their correlation with students' publication metrics, we seek to contribute to the growing body of literature on research productivity and to provide insights that can inform policy and practice in doctoral education and research policy. We also offer directions for future research around training graduate students for success in an era of global-mega science. At a time when publication metrics often serve as key indicators of

academic performance and career outcomes (Carpenter et al. 2014; Tregellas et al. 2018), our findings provide implications for individuals, institutions, and nations for personal development, career advancement, science production, and national competitiveness.

Literature Review

The PhD, originating in Germany, was introduced to North America in the 19th century. While relatively new to Canada compared to the U.S., the doctoral degree was widely available in Canadian universities by the 1960s (Elgar 2003). According to the most recent data from the Canadian Association for Graduate Studies (CAGS 2018), full-time doctoral enrollment in Canada reached approximately 50,000 students in 2015, with an additional 3,000 enrolled part-time. Disciplinary distributions show that physical and life sciences, social and behavioral sciences, and engineering collectively account for the majority of full-time PhD enrollments.

Canada's doctoral program places a strong emphasis on research, and publication activity is a central component of doctoral training (Larivière 2012). Publishing scientific research is one of the primary ways researchers establish scholarly reputations (Becher and Trowler 2001; Merton 1973). On one hand, faculty may choose to collaborate to pursue certain benefits, such as that collaborative work is often published in higher-quality outlets. On the other hand, when faculty collaborate, they are less productive as individuals (i.e., they publish less solo-authored work), and they do not receive full credit for co-authored publications (Bikard et al. 2015). Co-authored research is more likely to be cited and to have a longer "citation lifetime," which refers to a paper continuing to be cited over a longer period of time (Beaver 2004: 405). This is particularly important for PhD students in Canada, given that the main way they produce scientific publications is through collaboration (Larivière et al. 2006). A large-scale analysis of doctoral students in Quebec found that PhD students contributed to about a third of the province's publication output between 2000 and 2007, and the average number of authors of PhD students' papers ranged from 1.3 to 34 across disciplines (Larivière 2012). Positive relationships between publication activities, degree completion, and post-graduation research productivity were also found in the study.

International Research Collaboration

Internationalization plays a crucial role for university researchers to advance in their careers (Musselin 2004; Smeby and Trondal 2005). Just as with any research collaboration, IRC has benefits that make it attractive to individual researchers and challenges that prevent it from being more prevalent. For instance, IRC helps scholars transcend institutional and (national) cultural norms to make new discoveries. However, it also can require working through differences based on language, local reward systems, and career stage (Dusdal and Powell 2021). Despite potential challenges, IRC publications are more

frequently cited (e.g., Hsieh 2013). For example, Nguyen et al. (2017) found that, in Vietnam, multinational papers receive substantially more citations than those produced domestically.

In Europe, IRC has been prioritized as a metric for academic excellence and quality (Fox et al. 2017; Kwiek 2019; Mattsson et al. 2008). In multiple countries with relatively small higher education systems or academic markets, the academics who publish the most tend to participate in IRC (Kwiek 2016) and secure academic positions. Men seem to focus more and benefit more from such collaboration than women (Kwiek and Roszka 2021; Lindahl et al. 2021). A study of Danish academics found that women are less likely to participate in IRC; instead of publishing co-authored work in higher impact journals, they tend to publish more single-authored work in less frequently cited journals (Nielsen 2016). However, the disparity between men and women participating in IRC appears to be generational, and the gender gap disappears among early-career scholars (Kwiek & Roszka 2021). This gendered generational shift in IRC may reflect how women scientists often experience sexism, hostility, or exclusion by men in their home academic departments, so they look for international collaborators (Zippel 2017).

Compared to faculty, fewer studies examine IRC among graduate students. Scholars have confirmed a general trend of increasing doctoral student publishing over time, including through co-authored papers (Lee 2000). Some studies have examined doctoral student research output as a function of faculty mentorship (Dewett et al. 2005), support (Kim and Karau 2009), interactions (McGaskey 2015), or early career training (Austin 2002). These approaches are all important but often not uniform. For example, faculty advisors may help doctoral students improve their scholarly output by initiating frequent research discussions (Li and Fernandez 2024); some advisors encourage students to take greater responsibility in various research functions (Yoshioka-Kobayashi and Shibayama 2021). Without proper training and guidance from advisors, students are likely to have lower scientific production or even drop out of the program (Golde 2005).

For doctoral students, research collaborations are not only an opportunity to contribute to significant advancements in their fields but also a strategic way to establish a solid foundation for their future careers. The broader literature on research collaboration suggests that by working with scholars and researchers from different countries and institutions, doctoral students can access diverse perspectives, gain exposure to varied content knowledge and methodologies, and address global challenges with greater depth (Freshwater et al. 2006; Grathwol 2005; Jazvac-Martek 2009; Wagner et al. 2015; Yang et al. 2023). Moreover, the visibility and recognition from international co-authorship can significantly impact their professional trajectory, leading to more opportunities for funding, career development, and further research endeavors (Lindahl et al. 2021; Stoilescu and McDougall 2010). Therefore, understanding the dynamics of international collaboration and its influence on doctoral students' publication output is important for understanding how doctoral education can support early-career scholars seeking to establish academic impact.

Academic Mobility

Apart from IRC, some studies identify studying abroad as an important opportunity to gain international experience which is also linked to science production. For instance, Franzoni et al. (2011) found that researchers with international experience and mobility (e.g., earning a Ph.D. abroad) are more likely to engage in IRC and produce high-impact research. Their motivations are often beyond financial incentives, as these researchers are largely driven by the desire to establish and expand their research network. These foreign-born talents help the nation to maintain a high caliber of science production.

Though much of the research on study abroad examines undergraduate students, a few scholars have examined whether the benefits of international mobility extend to doctoral students, as well. Diogo et al. (2020) found that the opportunity to study abroad has been found to be positively related to international publication. One study examining Chinese doctoral students in STEM fields finds that 55% of Chinese PhD students participate in international collaborations and build networks during their time abroad, which in turn boosts their publication rates (Horta et al. 2021). Early international engagement not only benefits students' career prospects but also positively affects their advisors' research performance and institutional rankings, which led many universities to encourage doctoral students to publish internationally (Khalid et al. 2023). However, opportunities to study abroad are varied and may have differential influences. Eduan (2019) found that destinations and the depth of study experience are associated with international collaboration, while duration is not. Though these studies offer a few insights about the potential importance of study abroad for graduate students, more research is needed to understand whether opportunities to study abroad—independent of other opportunities for IRC—influence doctoral students' publication output.

Theoretical Framework: Intergroup Contact Theory

We draw upon intergroup contact theory to develop a set of hypotheses and interpret study findings. Allport's (1954) intergroup contact hypothesis states that meaningful interaction between members of different groups can occur when four optimal conditions are met, including equal status among participants, a common goal, cooperative (rather than competitive) collaboration, and support from authorities, law, or custom. Pettigrew (1998) later expanded this framework by adding a fifth condition, friendship potential, emphasizing the importance of context, time, and relationship depth to have meaningful contact.

Pettigrew's (1998) review of intergroup contact theory, identified an interesting, although not unique, dynamic: the theory holds practical and predictive utility even though most empirical studies that test Allport's (1954) hypothesis use cross-sectional data, do not account for self-selection, and do not reveal the causal mechanisms or processes that facilitate the benefits of intergroup contact. Even as Pettigrew (1998) added a potential condition for understanding how intergroup contact

can facilitate positive outcomes, he criticized that researchers can "overburden the hypothesis with facilitating, but not essential, conditions" (p. 65). Despite these limitations, Pettigrew (1998) argued that the hypothesis retained "policy importance" and "has proven useful in applied settings" (p. 66; see also Pettigrew 1971, 1975).

While access to international research collaboration in doctoral training may appear vague or ill-defined at first glance, we see how these opportunities can lead to positive outcomes through the lens of the intergroup contact theory. Opportunities to collaborate on research internationally may meet conditions for optimal contact. For instance, doctoral students and faculty advisors may share equal status as co-authors, share a common goal to publish their work, and work together (e.g., collecting/analyzing data and presenting findings) to achieve this goal. Institutional administrators, department heads, and funding agencies provide professional or financial support for these collaborations. Finally, recurring meetings provide the time and space for professional relationships (e.g., friendship potential) to form. This framework helps to explain why opportunities for international research collaboration (e.g., ongoing interactions that meet intergroup contact conditions) can influence the likelihood of doctoral students leading or contributing to publishable work. Although we do not test the causal mechanisms that underlie intergroup contact theory, following Pettigrew's (1998) review, we apply Allport's (1954) hypothesis to develop two hypotheses about how international opportunities can influence graduate students' research production:

H1: Graduate students' research output (having first or sole author publications, having co-authored with program faculty, the number of first or sole author publications) is positively related to the opportunity to collaborate on research internationally.

H2: Graduate students' research output (having first or sole author publications, having co-authored with program faculty, the number of first or sole author publications) is positively related to the opportunity to study abroad.

Data and Method

Data for this study were obtained from the Canadian Graduate and Professional Survey (CGPSS), which covers a broad array of graduate education metrics collected from 200,000 students across 50 Canadian universities. The CGPSS provides detailed information on various aspects of graduate student life, including demographic data, program quality, opportunities for research collaboration, self-reported publication activity, and interactions with advisors. Previous research has utilized CGPSS data to investigate graduate students' satisfaction levels (e.g., Spence 2009; Zhao 2012).

Records with missing values were excluded, and the dataset was filtered to include only responses from doctoral students in the year 2019, the most recent year of available CGPSS survey data. The final analytical sample consisted of 14,366 participants from ten distinct disciplines, categorized into two groups: "STEM fields" (Biological Sciences, Engineering, Environmental Sciences, Health Sciences, and Physical and Mathematical Sciences) and "Non-STEM fields" (Business/

Management, Education, Fine and Applied Arts, Humanities, and Social Sciences). Distinguishing between STEM and non-STEM fields is essential for examining discipline-specific differences in the effect of international collaboration and research outcomes (Xu 2013; Shin et al. 2016).

Variables

Three dependent variables were selected to assess graduate students' research output. The first variable, PRESPUB_AUTHOR, indicates whether respondents have published as the sole or first author in a refereed journal (1 = Yes; 0 = No). The second variable, PRESPUB_COAUTH, refers to whether respondents have published with faculties in their program (1 = Yes; 0 = No). The third variable, PRESPUB-NUM_AUTHOR, measures the number of sole or first author publications, ranging from "0" (no publications) to "4" (four or more publications).

Two key independent variables were chosen based on the study hypotheses. The first variable measures the extent of opportunities for IRC, with categories ranging from: 1=No opportunity; 2=Yes, to some extent; 3=Yes, to a great extent. The second variable assesses the opportunities for studying abroad, also categorized from 1 ("No opportunity") to 3 ("Yes, to a great extent").

Previous literature highlights several variables that may influence the relationship between IRC opportunities, study abroad experiences, and graduate students' publication rates. To ensure the accuracy of our results and conclusions, we included and coded control variables in our analysis. The first set of control variables pertains to students' demographics, such as age, gender, citizenship, parental status, academic discipline, and years of study. Specifically, we categorized age as follows: 1 = 20 or younger; 2 = 21-25; 3 = 26-30; 4 = 31-35; 5 = 36-40; 6 = 41-45; 7 = over 45. Gender was coded as 0 = Male; 1 = Female. These demographic factors are well documented in previous studies as strong predictors of doctoral students' academic experience (Epstein and Lachmann 2018; Roksa et al. 2022). Doctoral students' citizenship status was coded as 0 = Canadian Citizen or Permanent Resident; 1 = International, as it can affect access to financial support and research opportunities, which in turn influences their success in their Ph.D. journey (Feizi and Elgar 2023; Schneider et al. 2020). Additionally, we controlled for the number of children or dependents (0 = None; 1 = One child; 2 = Two children; 3 = Three children; 4 = Four or more children), given that family responsibility is correlated with doctoral students' well-being during graduate studies (Pocock et al. 2011). Academic discipline was recoded as 0 = Non-STEM and 1 = STEM to control for the fact that several studies highlight the differences in nature, funding, opportunities, and publication output between these STEM and non-STEM fields (Churchill et al. 2021; Borrego et al. 2021; Uddin et al. 2021). Finally, we included Years of Study, categorized as 1 = 1st year to 6 = 6th year or beyond, to account for doctoral students' current year of study in doctoral programs and its link to research output and publication (Khalid et al. 2023).

The second group of control variables pertains to the quality of the doctoral program, a critical predictor of doctoral students' research publication output (Khalid

et al. 2023). Several studies indicate the significance of having a supportive advisor (Dewett et al. 2005; Kim and Karau 2009), as they influence research skills development (Feldon et al. 2015), overall satisfaction (Trigwell and Dunbar-Goddet 2005), and doctoral degree persistence (Nettles and Millett 2006; Smith et al. 2006). The frequency of student-advisor interaction and advisors' availability were also found to be positively related to doctoral students' scholarly output (e.g., McGaskey 2015). Therefore, we utilized three index variables derived from the CGPSS data: Quality of Advisor, Quality of Teaching, and Research Training and Career Orientation. These indices were computed as the mean scores of responses to relevant questions. Quality of advisor includes questions like "My advisor gave me constructive feedback on my work (0 = Not applicable; 5 = Strongly agree)" and "My advisor was available for regular meetings (0 = Not applicable; 5 = Strongly agree)". For Quality of Teaching, items included assessments such as "The intellectual quality of the faculty (1 = Poor; 5 = Excellent)" and "Amount of coursework (1 = Poor; 5 = Excellent)". Research Training and Career Orientation included questions like "Advice/workshops on publishing your work (1 = Poor; 5 = Excellent)" and "Advice/workshops on professional ethics (1 = Poor; 5 = Excellent)".

The final set of control variables addressed doctoral students' internal motivation and career goals, which may influence their publication activity. We controlled for the importance students placed on various career objectives: (a) Work/collaborate with businesses, (b) Work/collaborate with not-for-profit organizations, and (c) Collaborate on research internationally, with ratings ranging from 1 = Not important to 3 = Very important. These variables control for the potential influence of students' motivations on their publishing apart from opportunities for IRC and study abroad. Descriptive statistics for both dependent and independent variables are presented in Table 1. We also provide a summary of how key variables varied between STEM and non-STEM fields in Table 2.

Method of Analysis

For the first two dependent variables, PRESPUB_AUTHOR and PRESPUB_COAUTH, binary logistic regression models were employed. These models regressed the binary outcomes on the two primary independent variables and the control variables. For the third dependent variable, PRESPUBNUM_AUTHOR, an ordinal logistic regression model was used. Although the response options are numeric, they represent an ordinal scale, with "4" representing four or more publications. Thus, ordinal logistic regression is appropriate for analyzing ordinal outcomes. The ordinal logistic regression yields a single set of coefficients based on the proportional odds assumption, which posits that the relationship between consecutive categories of the outcome variable (e.g., 0 publications versus 1 publication, 1 publication versus 2 publications) remains consistent across thresholds. However, a check of the proportional odds assumption using Harrell's (2023) graphical method showed variations in coefficients across different outcome categories. This indicates that the impact of international opportunities may differ across various levels of publication output.

Table 1 Descriptive statistics of CGPSS doctoral respondents, 2019 (N = 14,366)

Variables	Mean	SD	Min	Max
Dependent variables				
Published as sole or first author in a refereed journal	0.71	0.44	0	1
Co-authored in refereed journals with program faculty	0.64	0.48	0	1
Published as sole or first author in a refereed journal (numbers)	1.03	1.22	0	4
Independent variables of interest				
Opportunity for int'l research collaboration (IRC)	1.84	0.73	1	3
Opportunity for study abroad	1.68	0.69	1	3
Demographics				
Age	3.70	1.28	1	7
Gender (Woman)	0.54	0.50	0	1
Citizenship (International)	0.27	0.44	0	1
Children	0.70	0.94	0	5
Discipline (STEM)	0.59	0.49	0	1
Years of study	3.35	1.44	1	6
Program quality				
Advisor	3.36	0.64	1	4
Quality of teaching	3.78	0.85	1	5
Research training & career orientation	2.80	1.01	1	5
Importance				
Work/collaborate with businesses	1.99	0.82	1	3
Work/collaborate with not for profit organizations	1.97	0.79	1	3
Collaborate on research internationally	2.43	0.70	1	3

Variables with a max of "1" are binary variables (0=no; 1=yes). Other variables are generally ordinal variables with varied response scales: some are rated on a scale of 1 to 3, while others range from 0 to 5. Lower scores typically indicate poorer or lesser outcomes, whereas higher scores indicate more positive outcomes. Refer to the "Variables" for detailed explanations

Although we used ordinal regression for our preferred model to analyze the ordered count outcome variable, we also conducted a multinomial logistic regression analysis with PRESPUBNUM_AUTHOR to check for robustness. This regression model estimated the likelihood of being in each publication category (e.g., 1 publication, 2 publications) relative to a reference category (0 publications). This approach allows us to assess how international opportunities influence doctoral students' publications output across different levels. Additionally, in the follow-up analyses, we included interaction terms ("Opportunity for IRC * Woman", "Opportunity for IRC * STEM", "Opportunity for IRC * International students") to explore potential variation by gender, disciplines, and citizenship.

Table 2 Sample variation in by disciplines

	Number of Students	Had at Least One First or solo-Authored	Co-Authored with Program Faculty	Number of First or Solo-Authored Publications
Disciplines			*Mean*	
STEM				
Biological Science	1854	0.76	0.77	1.11
Engineering	2509	0.71	0.66	1.15
Environmental Science	529	0.66	0.62	0.92
Health Science	2261	0.78	0.75	1.32
Physical & Mathematics	1307	0.76	0.76	1.08
Non-STEM				
Business/Management	287	0.43	0.51	0.40
Education	896	0.67	0.62	0.79
Fine and Applied Arts	311	0.64	0.35	0.96
Humanities	1757	0.68	0.42	0.78
Social Sciences	2598	0.69	0.60	0.93
Others	57	0.63	0.53	0.74

Limitations

While the data used in this study enables us to explore the relationship between IRC and doctoral students' publication output, we acknowledge several limitations. First, although the CGPSS includes a large sample of graduate students in Canada, the results may not generalize to doctoral students in other countries. Second, the survey relied on self-reported data for several variables, which could introduce bias or subjectivity (Spector 1994). For example, control variables were self-reported with limited response options, which we further collapsed or recoded. One instance is the gender variable which included a third response option ("other"). This lacked clarity (i.e., it did not clarify whether the participant was indicating a non-binary identity or declining the binary male/female response option) and was selected by only a few participants. Further, the survey did not specify roles in career outcomes (work/collaborate with businesses; work/collaborate with not-for-profit organizations; collaborate on research internationally), which might affect the perceived importance of publishing.

With the use of secondary data from 2019, we recognize temporal limitations, particularly as the data were collected before the COVID-19 pandemic, which impacted scholarly publishing (Aviv-Reuven and Rosenfeld 2021). The opportunity of studying abroad is also likely to be affected by the pandemic's long-lasting effects on travel restrictions and international mobility. While using pre-COVID data could offer a more stable and consistent reference point for understanding the typical conditions and experiences associated with studying abroad, it might not account for the shifts and changes that have occurred since then.

Additionally, data collected in 2019 may not capture the significant changes in doctoral education, such as the development of different types of PhDs (e.g., university- and industry-based PhD) (Bröchner & Sezer 2020). Changes in doctoral education over past decades have influenced research and publication expectations. For instance, professional doctorate programs tend to emphasize applied research to improve practice, whereas research-focused PhD programs prioritize scholarly contributions in peer-reviewed journals (Neumann 2005). These differences in doctoral program types may also intersect with gender disparities, with prior work showing that women are underrepresented in academia and less published than their male peers (Lubienski et al. 2018). Similarly, institutional assessment criteria and career advancement opportunities vary across institutions and PhD types (Jones 2018). Traditional PhD programs prioritize research productivity in high-impact journals, while professional doctorates focus on enhancing practice (Belavy et al. 2020; Fulton et al. 2012). These differences in expectations for publication output and international collaboration could influence how research is evaluated and valued across doctoral programs. Future research should aspire to use longitudinal data to examine how changes in doctoral education (including how different types of doctoral programs value publishing) may influence graduate students' research output.

Moreover, the phrasing of survey items may cause confusion about whether publication counts represent an entire doctoral career or are limited to those from 2019. This issue is similar to previous work (e.g., Spence 2009), which also focused on data from specific years when examining doctoral students' outcomes. Additionally, if the opportunity to study abroad was assessed at a different time from when the research output was measured, there might be a disconnect between the perceived opportunity and actual experiences. Thus, the operationalization of the variable "opportunity" captures whether students had access to international collaboration or study abroad opportunities, rather than measuring actual participation. This distinction is important, as opportunities to engage in these experiences does not necessarily mean they participated in them. The independent variable "opportunity for study abroad" also does not differentiate between types, such as short-term exchanges versus long-term programs or specific destinations, which might obscure variations in how different opportunities impact publication outcomes. Although the temporal scope of this study may not fully capture all of doctoral students' academic output, it provides valuable insights into the relationship between international research opportunities and doctoral students' publication outcomes. Future research could use longitudinal data to understand publication productivity throughout doctoral programs by further exploring the impact of international research opportunities on scholarly output across different stages of graduate education.

Another notable limitation of this study is the absence of variables related to the role of doctoral advisors in providing or facilitating international opportunities, such as their international networks, mobility histories, and collaboration patterns, in this dataset. Studies have shown that advisors' professional networks and past experiences in international research may significantly shape students' access to IRCs and, consequently, their academic outcomes (Wang et al. 2010). Wang et al. (2021) found that doctoral supervisors often introduce Ph.D. candidates to their academic networks as a way to accelerate students' socialization process. Supervisors

with international collaboration histories or patterns may also help doctoral students acquainted with research elites internationally, shaping their academic outcomes. Additionally, Yoshioka-Kobayashi and Shibayama (2021) suggest that doctoral supervisors take a variety of approaches to helping students become autonomous researchers and that their training styles are influenced by their own experiences as doctoral students. We acknowledge that with the limits of secondary data, we were unable to consider how supervisors advise or help students network or become autonomous researchers. Nevertheless, the data allowed us to examine understudied opportunities for international engagement.

Additionally, the dataset did not specify which faculty members should be classified as "program faculty" for students' co-authored publications. PhD studies in Canada typically follow the German model—"research-based, master-apprentice-model" (Kehm 2006), where students are supervised by a primary advisor for research training. Presumably, "program faculty" refers to those directly involved in the supervision or academic training of PhD students within the program, such as students' primary advisors or supervisory committee members. However, we acknowledge that faculty roles and the structure of PhD programs can vary across universities and fields. As an example, the emergence of interdisciplinary graduate education, as evidenced by programs like the Canadian Institute of Health Research's Strategic Training Initiatives in Health Research (Vanstone et al. 2013), introduces cross-disciplinary co-supervision (Kiley 2009), where students receive joint supervision by faculties from multiple disciplines. The lack of specificity may complicate the interpretation of the data, as collaboration often extends beyond program boundaries. Future research may utilize other datasets that specify faculty roles (e.g., advisors, committee members, faculties from other departments) to distinguish program faculties and other collaborators and aim to incorporate these supervisor characteristics to explore how they influence students' experiences with IRCs. We recognized several limitations regarding disciplinary differences. First, we acknowledge that authorship conventions vary across disciplines, affecting interpretations of scholarly productivity. For instance, prior research shows that fields like mathematics, economics, and physics frequently list authors alphabetically while other disciplines list authors based on their contributions (Waltman 2012). In our study, we used first-authorship as a possible measure of research output Although we were working with de-identified data, which did not allow us to consider whether names were listed alphabetically, there is no reason to believe that data were systematically biased based on students' last names and would influence our analyses of the relationship between international opportunity and output. Nevertheless, Table 2 shows the means of key dependent variables and the distribution of students across majors, which provides a detailed overview of how scholarly output varied across disciplines in the dataset.

Additionally, we acknowledge that discipline differences are also reflected in publication norms. STEM fields, such as physics, tend to disseminate knowledge through articles (Bazerman 1988). STEM researchers prioritize publishing articles with a shorter publication period and higher volume. Conversely, publication norms are often not uniform across social science fields. For instance, humanities fields value publishing books and book chapters, while fields like economics resemble the

STEM model of article-based publishing (Nederhof, 2006; Savage & Olejniczak 2022). These differences may also influence students' priorities in publishing across different stages, majors, and institutions, as the requirement for earning a Ph.D. varies. In general, Canadian doctoral programs require some form of thesis/dissertation for Ph.D. degree earning, and the structure/format of a doctoral thesis is typically determined by students' supervisors and committee members (Saliba 2012). One format, the Publication-based Thesis (PBT), consists of a number of scholarly publications (usually three) focusing on the same research problem. These publications can be co-authored, as long as students' contribution is considered substantial (University of Toronto, 2020). With secondary data, we were unable to account for these nuanced differences in publication requirement and authorship, Future researchers may include these variations when collecting data for more accurate, field-sensitive interpretations.

Finally, due to the evolving nature of academic research, where international collaboration has become an embedded feature of the research process itself (Wagner et al. 2015), we recognize that the relationship between the opportunity of international research collaboration and publication outputs is correlational rather than causal. Future research could use different datasets and more advanced statistical methods to assert causal inferences.

Findings

Table 3 provides a summary of the models analyzed in this study, highlighting key variables and their effects on publication outcomes among graduate students. For the first hypothesis (H1: Graduate students' research output (having first or sole author publications, co-authored with program faculty, the number of first or sole author publications) is positively related to the opportunity to collaborate on research internationally. We found a positive, statistically significant relationship between the opportunity for IRC and all outcomes. Specifically, when both were compared to the reference category (no opportunity), doctoral students with extensive IRC opportunities exhibited larger estimated effects than those with moderate levels of collaboration opportunities across all three publication outcomes. As an example, for doctoral students with extensive international IRC opportunities, the log odds of having a first or sole author publication ($\beta = 0.61$, $p < 0.001$) is approximately 2.5 times higher compared to those with moderate collaboration opportunities ($\beta = 0.25$, $p < 0.001$); Furthermore, the log odds of publishing a higher number of first or sole author publications ($\beta = 0.57$, $p < 0.001$) is approximately twice as high for those with extensive collaboration opportunities relative to those with moderate opportunities ($\beta = 0.25$, $p < 0.001$).

In contrast, for the second hypothesis (H2: Graduate students' research output (having first or sole author publications, co-authored with program faculty, the number of first or sole author publications), we found more complex results. The opportunity for study abroad appears to be positively, although relatively small, related to having at least one first or sole-authored publication ($\beta = 0.09$, $p < 0.05$), yet negatively related to having co-authored with program faculty. For example, doctoral

Table 3 Summary of results for regression models examining graduate students' research publications

	Had at Least One First or Solo-Authored		Co-Authored with Program Faculty		Number of First or Solo-Authored Publications	
	Log odds	SE	Log odds	SE	Log odds	SE
Opportunity for international research collaboration (IRC)						
To some extent	0.25***	0.05	0.43***	0.04	0.25***	0.04
To a great extent	0.61***	0.07	0.83***	0.06	0.57***	0.05
Opportunity for study abroad						
To some extent	0.09*	0.05	– 0.10*	0.04	0.01	0.04
To a great extent	0.001	0.07	– 0.32***	0.07	– 0.003	0.06
Demographics						
Age	– 0.32***	0.02	– 0.20***	0.02	– 0.24***	0.02
Gender (Woman)	0.11	0.04	0.16***	0.04	– 0.13***	0.03
Children	0.71***	0.03	0.22***	0.02	0.40***	0.02
Citizenship (International)	– 0.31***	0.05	– 0.51***	0.04	– 0.41***	0.04
Discipline (STEM)	0.43***	0.04	0.81***	0.04	0.49***	0.04
Years of study	0.27***	0.01	0.16***	0.01	0.40***	0.01
Program quality						
Advisor	0.23***	0.04	0.23***	0.03	0.28***	0.03
Quality of teaching	– 0.09***	0.03	– 0.10***	0.03	– 0.21***	0.02
Research training & career orientation	0.03	0.02	0.12***	0.02	0.03	0.02
Goals						
Work/collaborate with businesses	– 0.18***	0.03	0.01	0.03	– 0.11***	0.02
Work/collaborate with not for profit organizations	– 0.09**	0.03	– 0.04	0.03	– 0.01	0.02
Collaborate on research internationally	0.05	0.03	– 0.02	0.03	0.16***	0.03

*$p \leq 0.05$, **$p \leq 0.01$, ***$p \leq 0.001$

students with extensive study abroad opportunities ($\beta = -0.32$, $p < 0.001$) are significantly less likely to co-author with program faculty than those with no study abroad opportunities. Lastly, we found non-statistically significant relationships between the opportunity to study abroad and the *number* of first or sole-authored publications.

Robustness Check

The result from ordinal logistic regression shows that opportunity for IRC is positively related to various outcomes of doctoral students' publication output. To delve deeper into this relationship, we conducted multinomial logistic regression analyses. The model use "having 0 publications" as the baseline category to assess how independent variables related to the likelihood of doctoral students achieving different

Table 4 Results of Multinomial regression models examining graduate students' numbers of first or sole author publications. Reference level: 0 publication

	One publication		Two publications		Three publications		Four and more publications	
	Log odds	SE	Log odds	SE	Log odds	SE	Log odds	SE
Opportunities for international research collaboration (IRC)								
To some extent	0.30***	0.05	0.29***	0.007	0.25**	0.09	0.40***	0.09
To a great extent	0.55***	0.07	0.66***	0.09	0.63***	0.12	0.99***	0.12
Opportunities for study abroad								
To some extent	– 0.04	0.05	– 0.03	0.06	0.10	0.08	0.03	0.08
To a great extent	– 0.09	0.08	0.07	0.10	– 0.07	0.13	– 0.01	0.12
Demographics								
Age	– 0.27***	0.02	– 0.31***	0.03	– 0.38***	0.04	– 0.29***	0.04
Gender (Woman)	0.10***	0.04	– 0.09***	0.06	– 0.11***	0.07	– 0.46***	0.07
Children	0.51***	0.03	0.56***	0.03	0.55***	0.04	0.59***	0.04
Citizenship (International)	– 0.32***	0.05	– 0.44***	0.07	– 0.58***	0.09	– 0.57***	0.09
Discipline (STEM)	0.35***	0.05	0.50***	0.06	0.76***	0.09	0.93***	0.09
Years of study	0.31***	0.01	0.49***	0.02	0.60***	0.03	0.61***	0.03
Program quality								
Advisor	0.24***	0.04	0.37***	0.05	0.38***	0.07	0.41***	0.07
Quality of teaching	– 0.14***	0.03	– 0.31***	0.04	– 0.30***	0.05	– 0.30***	0.05
Research training & career orientation	– 0.01	0.03	0.05	0.03	0.08	0.04	0.02	0.04
Goals								
Work/collaborate with businesses	– 0.12***	0.03	– 0.16***	0.04	– 0.16**	0.05	– 0.16**	0.05
Work/collaborate with not for profit organizations	– 0.05	0.03	– 0.01	0.04	0.04	0.05	0.01	0.05
Collaborate on research internationally	0.13***	0.04	0.15**	0.05	0.31***	0.06	0.28***	0.06

*$p \leq 0.05$, **$p \leq 0.01$, ***$p \leq 0.001$

levels of publications (e.g., one publication, two publications, etc.). The results from these analyses are shown in Table 4.

The multinomial logistic regression results demonstrate a strong, positive association between IRC opportunities and the likelihood of achieving higher publication counts, which align closely with the estimates in Table 3. The magnitude of impact is significantly greater for those with more extensive opportunities. The impact of extensive IRC opportunities is significantly pronounced. Students with substantial opportunities for IRC show markedly higher log odds values across all categories, compared to the reference group. As an example, the odds of a student achieving 4 or more publications (compared to having no publications) are 2.69 (e^0.99 ≈ 2.69) times higher when they have a great extent of IRC opportunities.

Additional Analyses

Variation by Gender

As discussed earlier in this study, existing literature has highlighted significant gender disparities in publication output and IRC (Kwiek and Roszka 2021). For doctoral students across disciplines, men tend to have more publications and scholarly output than women (Lubienski et al. 2018). Within our aggregated sample, we found a negative correlation between being a woman and the likelihood of publishing two or more first or sole authored articles (refer to Table 3), which align with previous studies. Additionally, while it is not clear whether the availability of IRC differ by gender, women are less likely to participate in IRC due to their lower international mobility, especially for those at advanced career stages and older age(Kwiek and Roszka 2021; Jöns 2011).

To further examine whether gender influences extend to the relationship between doctoral students' IRC opportunities and publication output, we added an interaction term, "Opportunity for IRC * Woman," into both ordinal logistic regression models. However, this interaction term did not reach statistical significance (see Table 5). This finding indicates that, in our sample, although women are independently less likely to publish two or more first or sole authored articles, opportunity for IRC does not differentially impact their publication output compared to men.

Variation between STEM and Non-STEM Fields

Across our pooled analysis, we found that majoring in STEM disciplines was positively associated with the likelihood of having a first or sole-author publication and higher publication outputs (see Table 3). To further explore discipline-based difference and its influence on our key variables, we added the interaction term Opportunity for IRC * STEM into our models. The interaction term for having IRC opportunities "to some extent" and STEM field major shows non-significant estimates across all outcomes, which suggests that the effect of having some extent of IRC opportunities does not significantly differ for STEM versus non-STEM fields.

Table 5 Summary of results for regression models examining graduate students' research publications (with interaction terms)

	Had at Least One First or Solo-Authored		Co-Authored with Program Faculty		Number of First or Solo-Authored Publications	
	Log odds	SE	Log odds	SE	Log odds	SE
Opportunity for international research collaboration (IRC)						
To some extent	0.23**	0.08	0.47***	0.08	0.30***	0.08
To a great extent	0.72***	0.14	1.01***	0.12	0.81***	0.10
Opportunity for study abroad						
To some extent	0.10*	0.05	– 0.10*	0.04	0.02	0.04
To a great extent	0.02	0.07	– 0.31***	0.07	0.02	0.06
Demographics						
Age	– 0.32***	0.02	0.20***	0.02	– 0.24***	0.02
Gender (Woman)	0.05	0.06	0.20**	0.06	– 0.17**	0.06
Children	0.71***	0.03	0.22***	0.02	0.40***	0.02
Citizenship (International)	– 0.11	0.08	– 0.38***	0.07	– 0.29***	0.07
Discipline (STEM)	0.42***	0.07	0.80***	0.06	0.59***	0.06
Years of study	0.27***	0.01	0.16***	0.01	0.40***	0.01
Program quality						
Advisor	0.23***	0.04	0.23***	0.03	0.28***	0.03
Quality of teaching	– 0.09**	0.03	0.10***	0.03	– 0.20***	0.02
Research training & career orientation	0.03	0.02	0.12***	0.02	0.03	0.02
Goals						
Work/collaborate with businesses	– 0.18***	0.03	0.01	0.03	– 0.11***	0.02
Work/collaborate with not for profit organizations	– 0.09**	0.03	– 0.04	0.03	– 0.01	0.02
Collaborate on research internationally	0.04	0.03	– 0.03	0.03	0.15***	0.03
Interaction terms						
Opportunity of international research collaboration						
To some extent * woman	0.11	0.09	– 0.04	0.08	0.05	0.07
To a great extent * woman	0.06	0.12	– 0.13	0.11	0.03	0.09
To some extent * STEM	0.04	0.09	0.06	0.08	– 0.10	0.08
To a great extent * STEM	– 0.15	0.13	– 0.07	0.12	– 0.28***	0.10
To some extent * International students	– 0.28**	0.10	– 0.19	0.10	– 0.11	0.09
To a great extent * International students	– 0.35**	0.13	– 0.21	0.12	– 0.27**	0.10

*$p \leq 0.05$, **$p \leq 0.01$, ***$p \leq 0.001$

However, when graduate students reported having IRC opportunities to "a great extent," the interaction term was statistically significant ($\beta = -0.28$, $p < 0.001$) for the outcome that measured the number of first or sole-authored publications. This significant negative coefficient implies that, compared to non-STEM major students, the beneficial effect of extensive IRC opportunities is substantially diminished

for individuals in STEM fields (not that they incur penalties by having more opportunities).

Variation by International Student Status

Findings from Table 3 suggested that international students were less likely to have publication output across all three outcomes. Our last interaction term examined whether the relationship between opportunities to study abroad and doctoral students' publication output varied by citizenship. We found that, compared to local students, international students might benefit less from IRC opportunities in terms of first-author publications. The estimates for the interaction term were negative for both the likelihood of having at least one first or solo-authored paper and the number of first or solo-authored publications outcomes when international students shared that IRC opportunities were supported "to a great extent."

Discussion and Conclusion

This study examined how the opportunities for international engagement, such as opportunities to collaborate on research and to study abroad influence doctoral students' publication output. Publishing in graduate school is an important outcome for doctoral students' scholarly development and career opportunities (e.g., Becher and Trowler 2001). Additionally, doctoral students are an important, yet understudied, source of human capital for countries to sustain or increase overall research production in an era of mega-global science (Baker & Powell 2024).

Our analyses of data from the CPGSS sheds light on the importance of supporting IRC opportunities as a part of doctoral training, and potentially a key driver of research productivity among doctoral students. For our first hypothesis, we found that having greater opportunities for IRC was positively related to all three publication outputs (publishing at least once as an independent or first author, publishing with program faculty, and number of first- or solo-authored publications). The positive relationship between the opportunities of IRC and doctoral students' first or sole-author publication adds an important dimension to the existing literature on research collaboration. These results suggest that IRC not only helps students increase overall publication output, but it also provides opportunities for doctoral students to take leading roles in research projects.

The distinction between overall publication output and first-author publications is significant in the context of doctoral education and early-career researcher development. While increased co-authorship opportunities are valuable, first-author publications carry particular weight for doctoral students. For doctoral students entering the job market, specifically those looking for faculties and researcher positions, having first-author publications can significantly enhance their competitiveness. In many academic fields, first-author publications are given more weight in hiring, tenure, and promotion decisions, as it is used as a proxy for the level of contribution in many disciplines (Hagen 2010).

Based on the importance of publishing as lead author or solo-authored work for graduate students' longer-term career success, it was important to consider whether certain groups of students were more or less likely to benefit from IRC. Prior research indicated inconsistent findings about whether women were less likely than men to benefit from IRC (e.g., Kwiek and Roszka 2021; Lindahl et al. 2021; Nielsen 2016; Zippel 2017). We found that that women did not benefit more or less than men from opportunity for IRC. Recall that Kwiek and Roszka (2021) concluded that the gender gap in IRC may have been attributed to a generational gap, whereby early- career women were no longer less likely to collaborate internationally than early- career men. Because our analyses were limited to graduate students, it is not surprising that we did not find evidence of heterogenous effects on the benefits of IRC between men and women.

When we examined potential differences between academic fields (i.e., STEM and non-STEM), our results revealed diminishing marginal returns to IRC opportunities among STEM students. While extensive IRC opportunities may generally enhance research outcomes, the effect is less pronounced for those in STEM disciplines, potentially due to field-specific factors such as already having higher funding (Borrego et al. 2021), more resources (Behrens and Gray 2001), and high levels of interdisciplinary research collaboration (Uddin et al. 2021) in STEM. Non-STEM students, who may have less exposure to research environments and collaborative opportunities compared to their STEM counterparts are likely to benefit more from IRC opportunities. It is also possible that greater opportunities for IRC cannot overcome that STEM graduate students are reliant on local laboratory resources or materials and data tied to their local faculty supervisors.

Finally, our results suggested that, compared to Canadian-born doctoral students, international doctoral students tended to have lower publication output (refer to Table 3). Our follow-up analyses with interaction terms also suggested that international doctoral students were also less likely to benefit from IRC opportunities. This issue is particularly relevant in the context of higher education where universities globally compete to attract and retain international students (Douglass and Edelstein 2009). While international students contribute substantially to the revenue and international profile of HEIs, our analysis suggests that institutions may not be adequately supporting these students in a way that maximizes their research output. This finding aligns with some aspects of earlier literature that found no effect on geographical mobility in terms of collaboration outputs (Bernela and Milard 2016) but also contradicted more recent studies by Horta et al. (2021), in which they found that more than half of the doctoral students established research collaboration within hosting universities during their time abroad.

Although outside the direct scope of this paper, we offer several possible explanations. First, international students might face more language and cultural barriers that impede their progress in scholarly writing and publishing (Dusdal and Powell 2021). They may also experience various physical and mental challenges during their transition to graduate education (Evans and Stevenson 2010). Second, local students may be more familiar with local research environments and culture, such as research priorities and publishing norms, which may give them advantages that lead to publication. Third, their interaction with advisors

may be obstructed because of cultural differences. For example, Doutrich (2001) found that some international students may hesitate to reach out to their advisors. This self-withdrawal could be detrimental to their scholarly output since the frequency of having research discussions is positively correlated with the number of publications (McGaskey 2015). Finally, it is possible that, as with STEM students, the negative estimates were evidence of diminishing returns. Many international students already have international relationships, and therefore, they may not receive the same benefit from having IRC opportunities as domestic graduate students.

We found mixed results for the second hypothesis that focused on study abroad opportunities. Having greater study abroad opportunities had a positive, though practically small influence on graduate students' odds of ever having published as a sole or first author. However, it had negative influence on co-authoring with faculty. Study abroad opportunity did not have a statistically significant relationship with overall publication output.

This finding is consistent with previous literature on the substantial benefits of IRC, as they provide more resources, access, diverse perspectives, and scientific networks to doctoral students (Freshwater et al. 2006). Doctoral students with extensive study abroad opportunities may spend significant time away from their home institution or are more inclined to collaborate with researchers they meet abroad. However, this physical separation and shift in focus may result in limiting their interactions and collaborations with local faculty, leading to fewer co-authorships with program faculty. The null results on overall productivity may reflect the small positive effect on publishing at least once as a first or solo author and the negative effect on publishing with faculty.

We acknowledge that our findings regarding study abroad are limited in that the variable does not indicate whether graduate students actually participated in study abroad. The mere opportunity to study abroad does not guarantee that students will take full advantage of it or that they will experience the benefits typically associated with such opportunities. Secondly, actual participation in study abroad programs might involve various challenges, such as difficulties in languages and adapting to new research environments (Edwards and Ye 2018), which could temporarily impact research productivity. The effectiveness of studying abroad can vary greatly depending on the program's quality, focus, and duration. For instance, some study abroad programs have a short duration, and the short duration might limit the ability of students to fully capitalize on new experiences and professional networks to enhance their research output (DeLoach et al. 2021). Lastly, doctoral students' research efforts could be detracted by the psychological and emotional aspects of studying abroad, including stress related to being away from familiar support systems or lack of preparation (Waheed 2022). Thus, while the opportunity to study abroad may associated with potential benefits, its direct impact on doctoral students' research output is less pronounced, possibly due to the variability in how these opportunities are utilized and experienced.

Implications

The positive relationship between IRC opportunities and doctoral students' first or sole authored publication output has important meaning for the field of academic research and doctoral education. For doctoral students, the opportunities of IRC could serve as a potential strategic mechanism for the development of independent research capabilities. The research skills and interdisciplinary thinking that develop through international collaborations are increasingly valuable for academic careers in a globalized research landscape (White 2019). Students with a research-related career goal are encouraged to actively participate in IRC and seize such opportunities to potentially improve their future competitiveness in the academic job market (Tregellas et al. 2018).

At the institutional levels, educational leaders and institutional personnel may consider strategies to optimize opportunities for international collaboration for students with the goal of enhancing publication output and productivity. For instance, institutions could build and maintain strategic partnerships with universities and research institutions around the world, as well as establish collaborative programs where such partnerships or collaborations are not already in place. Participation in international collaborative research projects, such as faculties exchange and foreign study, has become mainstream among highly selective colleges and universities in multiple countries (Sakamoto and Chapman 2011). Additionally, institutions should review their internationalization strategies to ensure that international students are not only offered equitable access to the resources and opportunities for collaborative research but also positioned to take full advantage of these opportunities. Universities may support these students through additional guidance, such as language support, accessing funding opportunities, and expanding academic networks.

In addition, faculties who participated in IRC are encouraged to introduce their international research networks to students, which could help students connect with a broader range of researchers globally, leading to increased research productivity. Prior literature indicated that, when returning to their home countries, researchers often continue to collaborate with their peers in graduate school, which makes research a "borderless activity" (Sakamoto and Chapman 2011: 6). Their foreign network could help expand the knowledge base and broaden research area, which build research capacity for local institutions and researchers (Kuzhabekova and Lee 2018). For doctoral students, the faculty's academic network could serve as a valuable resource, since it creates opportunities for mentorship, collaboration, and knowledge exchange beyond the home institution (e.g., Horta et al. 2021).

To enhance research production among doctoral students, policymakers could focus on developing and strengthening endeavors that facilitate and incentivize international collaborations. Implementing targeted funding schemes and establishing bilateral agreements between countries could expand opportunities for joint research initiatives, which often result in the "creation of branch campuses and joint research and technology initiatives" (Sakamoto and Chapman 2011: 4). More and more of these programs are breaking the historical dominance of Western countries with an increasing number of international collaborations emerging between non-Western partners, such as joint initiatives between China and Japan (Huang 2010). These

initiatives would enable doctoral students to engage more readily in international research, leading to scientific advancement. In this globalized world, IRC opportunities are required to solve global problems (The Royal Society 2011). As a growing force for scientific progress, the participation of doctoral students in international cooperation contributes to the globalization of knowledge production, accelerating the dissemination and impact of scientific discoveries.

Directions for Future Research

As an alternative way to examining doctoral students' self-reported publications output, future research can use bibliometric or scientometric data to examine publication quality and author affiliation. Investigating whether different international opportunities contribute to not only a higher volume of research outputs but also to more impactful and higher-quality publications—measured by citation rates, journal impact factors, and peer reviews—may offer a deeper understanding of their benefits. In both the U.S. and Canada, the quality of papers, measured by the Journal Impact factor, has been used in the review, promotion, and tenure process in most research-intensive institutions (McKiernan et al. 2019). This dual focus could help determine if these factors lead to both increased research productivity and enhanced research excellence.

Additionally, future research could measure actual participation in international collaboration or study abroad opportunities to understand their effects on doctoral students' research outputs. While this study explores whether students had access to these opportunities, future studies could explore the effects of participating in international co-authorships, institutional affiliations, or time abroad. This distinction could shed light on differences in scholarly outcomes associated with access to opportunities compared to participation. For instance, scholars may study how doctoral students learn about opportunities for international engagement. For instance, prior literature suggests that supervisors help facilitate international opportunities (e.g., Wang et al. 2010, 2021) and become more autonomous researchers (Yoshioka-Kobayahshi and Shibayama 2021), and it would help to understand how graduate students may benefit from advisors' networking and publishing. However, it is important to recognize that supervisor-student relationships can differ significantly depending on academic culture and institutional contexts. Research has documented instances of power imbalances in these relationships. For example, Larivière's (2012) study of Quebec doctoral students revealed that while most of their publications were co-authored, these collaborations predominantly involved supervisors or local research teams rather than international collaborators—a pattern distinct from the broader trends in Quebec academia. Similarly, Niemczyk's (2019) qualitative research explored the power dynamics faced by doctoral students working as research assistants. Some students reported feeling vulnerable in their relationships with advisors, which impedes their collaborative participation and long-term academic trajectories. Therefore, future research may examine how graduate students participate in international research collaborations (i.e., virtually or in-person) and seek to identify factors and venues (e.g., visas, conferences, travel grants) that can

facilitate further collaboration. Additionally, future research should adopt a cross-cultural perspective when examining the impact of advisor support and collaboration opportunity, as well as consider power dynamics and potential constraints of supervisory relationships.

We used intergroup contact theory (Allport 1954; Pettigrew 1998) as a theoretical basis for interpreting our findings about why access to international research collaboration can positively influence the research outputs of doctoral students. Pettigrew (1998) explained that Allport's (1954) hypothesis consistently served as a practical framework for empirical research even when studies are unable to test multiple conditions for intergroup contact or the processes that facilitate positive outcomes. Because this study relied on secondary data, we did not have information on the frequency, quality, or substance of interactions. Even if we had access to more detailed survey responses, it would be challenging to standardize participation in opportunities across thousands of responses. For example, even if we could measure the duration of interactions, it is difficult to systematically and consistently collect data from doctoral students on the substance of those interactions. However, guided by intergroup contact theory, prior literature shows that ongoing, supportive interactions where participants work toward a shared goal contributes to positive interprofessional relationships and outcomes (Bridges and Tomkowiak 2010; Mohaupt et al. 2012). Intergroup contact theory also helps explain why study abroad opportunities were not as statistically significant in this study, as these opportunities place less emphasis on building and maintaining professional relationships.

In this study, we used data from CGPSS 2019, and we recognized temporal limitations. One way to understand the long-term effects of international opportunities on doctoral students' publication output is by utilizing longitudinal studies. Tracking doctoral students from the beginning to the end of their programs could help in understanding the development of these effects, such as how early versus late-stage doctoral students are influenced differently by IRC opportunities (Horta and Santos 2016). Additionally, longitudinal data could highlight the sustainability of collaborative benefits, providing a clearer picture of how ongoing international engagement influences long-term scholarly productivity and career development (Kushkowski et al. 2003).

Future research could also explore how the impact of international opportunities varies across different disciplines and regions. As mentioned earlier, there is a great deal of differences between STEM and non-STEM majors (e.g., Behrens and Gray 2001; Borrego et al. 2021). Besides the nature of these disciplines, the type of publication—book, journal, etc.—differs as well (Savage and Olejniczak 2022). Similarly, regional differences in research collaborative practices could influence the effectiveness of international partnerships (Pohl 2020; Onyancha 2021). Future research may conduct comparative studies to further examine how the effects are shaped by contextual factors.

References

Allport, G. W. 1954. *The nature of prejudice*. Reading/Addison-Wesley.

Austin, Ann E. 2002. Preparing the next generation of faculty: Graduate school as socialization to the academic career. *The Journal of Higher Education* 73(1): 94–122. https://doi.org/10.1080/00221546.2002.11777132.

Baker, David P., and Justin J.W. Powell. 2024. *Global mega-science: Universities, research collaborations, and knowledge production*. Stanford: Stanford University Press.

Beaver, Donald. 2004. Does collaborative research have greater epistemic authority? *Scientometrics* 60(3): 399–408. https://doi.org/10.1023/B:SCIE.0000034382.85360.cd.

Becher, Tony, and Paul Trowler. 2001. *Academic tribes and territories*. Philadelphia: McGraw-Hill Education.

Behrens, Teresa R., and Denis O. Gray. 2001. Unintended consequences of cooperative research: impact of industry sponsorship on climate for academic freedom and other graduate student outcome. *Research Policy* 30(2): 179–199. https://doi.org/10.1016/S0048-7333(99)00112-2.

Belavy, Daniel L., Patrick J. Owen, and Patricia M. Livingston. 2020. Do successful PhD outcomes reflect the research environment rather than academic ability? *PLoS ONE* 15(8): 1–14. https://doi.org/10.1371/journal.pone.0236327.

Bernela, Bastien, and Béatrice. Milard. 2016. Co-authorship network dynamics and geographical trajectories-what part does mobility play? *Bulletin of Sociological Methodology/bulletin De Méthodologie Sociologique* 131(1): 5–24. https://doi.org/10.1177/0759106316642705.

Bikard, Michaël, Fiona Murray, and Joshua S. Gans. 2015. Exploring trade-offs in the organization of scientific work: Collaboration and scientific reward. *Management Science* 61(7): 1473–1495. https://doi.org/10.1287/mnsc.2014.2052.

Borrego, Maura, Nathan Hyungsok Choe, Kevin Nguyen, and David B. Knight. 2021. STEM doctoral student agency regarding funding. *Studies in Higher Education* 46(4): 737–749. https://doi.org/10.1080/03075079.2019.1650737.

Bridges, D. R., and J. Tomkowiak. 2010. Allport's intergroup contact theory as a theoretical base for impacting student attitudes in interprofessional education. *Journal of Allied Health* 39(1): 29–33.

Bröchner, Jan, and Ahmet Anil Sezer. 2020. Effects of construction industry support for PhD projects: The case of a Swedish scheme. *Industry and Higher Education* 34(6): 391–400. https://doi.org/10.1177/0950422220904932.

Canadian Association for Graduate Studies [CAGS]. 2018. 44th Statistical Report, 2018 Part I. Available at: https://cags.ca/wp-content/uploads/2019/11/2019-44th-Statistical-Report-Part-1-Englishoct2.pdf

Carpenter, Christopher R., David C. Cone, and Cathy C. Sarli. 2014. Using publication metrics to highlight academic productivity and research impact. *Academic Emergency Medicine* 21(10): 1160–1172. https://doi.org/10.1111/acem.12482.

Chen, Kaihua, Yi Zhang, and Fu Xiaolan. 2019. International research collaboration: An emerging domain of innovation studies? *Research Policy* 48(1): 149–168. https://doi.org/10.1016/j.respol.2018.08.005.

Churchill, Meryl P., Daniel Lindsay, Diana H. Mendez, Melissa Crowe, Nicholas Emtage, and Rhondda Jones. 2021. Does publishing during the doctorate influence completion time? A quantitative study of doctoral candidates in Australia. *International Journal of Doctoral Studies* 16:689–713. https://doi.org/10.28945/4875.

de Solla Price, Derek J. 1961. *Science since Babylon*. New Haven: Yale University Press.

de Solla Price, Derek J. 1963. *Little science, big science, and beyond*. New York: Columbia University Press.

DeLoach, Stephen B., Mark R. Kurt, and Neal H. Olitsky. 2021. Duration matters: Separating the impact of depth and duration in study abroad programs. *Journal of Studies in International Education* 25(1): 100–118. https://doi.org/10.1177/1028315319887389.

Dewett, Todd, Shung Jae Shin, Soo Min Toh, and Matthew Semadeni. 2005. Doctoral student research as a creative endeavour. *College Quarterly 8*(1). http://www.senecac.on.ca/quarterly/2005-vol08-num01-winter/dewett_shin_toh_semadeni.html

Diogo, Sara, Ricardo Biscaia, and Teresa Carvalho. 2020. Does taking a PhD abroad increase the number of international publications? Reflection from the best self-reported publication in Portuguese academia. *INTED2020 Proceedings*: 3428-3440. https://doi.org/10.21125/inted.2020.0975

Doutrich, Dawn. 2001. Experiences of Japanese nurse scholars: Insights for US faculty. *Journal of Nursing Education* 40(5): 210–216. https://doi.org/10.3928/0148-4834-20010501-06.

Douglass, John Aubrey, and Richard Edelstein. 2009. The Global Competition For Talent The Rapidly Changing Market for International Students and the Need for a Strategic Approach in the US.

Dusdal, Jennifer, and Justin J.W. Powell. 2021. Benefits, motivations, and challenges of international collaborative research: A sociology of science case study. *Science and Public Policy* 48(2): 235–245. https://doi.org/10.1093/scipol/scab010.

Eduan, Wilson. 2019. Influence of study abroad factors on international research collaboration: Evidence from higher education academics in sub-Saharan Africa. *Studies in Higher Education* 44(4): 774–785. https://doi.org/10.1080/03075079.2017.1401060.

Edwards, Viv, and Ye Ye. 2018. A sociological exploration of the impact of study abroad on international PhD students' self-identity. *Journal of Intercultural Communication* 18(2): 1–15. https://doi.org/10.36923/jicc.v18i2.757.

Elgar, Frank J. 2003. PhD degree completion in Canadian universities. *Nova Scotia, Canada: Dalhousie University:* 1–31.

Epstein, Nurith, and Daniel Lachmann. 2018. Gender inequity during the Ph.D.: Females in the life sciences benefit less from their integration into the scientific community. *Social Sciences* 7(8): 140. https://doi.org/10.3390/socsci7080140.

Evans, Catrin, and Keith Stevenson. 2010. The learning experiences of international doctoral students with particular reference to nursing students: A literature review. *International Journal of Nursing Studies* 47(2): 239–250. https://doi.org/10.1016/j.ijnurstu.2009.05.025.

Feizi, Samira, and Frank Elgar. 2023. Satisfaction, research productivity, and socialization in doctoral students: Do teaching assistantship, research assistantship and the advisory relationship play a role? *Heliyon* 9(9): e19332. https://doi.org/10.1016/j.heliyon.2023.e19332.

Feldon, David F., Michelle A. Maher, Melissa Hurst, and Briana Timmerman. 2015. Faculty mentors', graduate students', and performance-based assessments of students' research skill development. *American Educational Research Journal* 52(2): 334–370. https://doi.org/10.3102/0002831214549449.

Fernandez, Frank, & David P. Baker. 2017. Science production in the United States: An unexpected synergy between mass higher education and the super research university. In *The century of science: The global triumph of the research university*, eds. Justin J.W. Powell, David P. Baker, & Frank Fernandez, 85–111. Bingley: Emerald.

Fernandez, Frank, David P. Baker, Fu Yuan Chih, Ismael G. Munoz, and Karly S. Ford. 2021. A symbiosis of access: Mass undergraduate education, decentralization, flexible mission charters, and proliferating STEM PhD training in the U.S. from 1920–2010. *Minerva* 59(1): 79–98. https://doi.org/10.1007/s11024-020-09422-5.

Fox, Mary Frank, Mary Lynn Realff, Diana Roldan Rueda, and Jillian Morn. 2017. International research collaboration among women engineers: frequency and perceived barriers, by regions. *Journal of Technology Transfer* 42(6): 1292–1306. https://doi.org/10.1007/s10961-016-9512-5.

Franzoni, Chiara, Giuseppe Scellato, and Paula Stephan. 2011. Changing incentives to publish. *Science* 333(6043): 702–703. https://doi.org/10.1126/science.1197286.

Freshwater, Dawn, Gwen Sherwood, and Vicki Drury. 2006. International research collaboration: Issues, benefits and challenges of the global network. *Journal of Research in Nursing* 11(4): 295–303. https://doi.org/10.1177/1744987106066304.

Fulton, John, Judith Kuit, Gail Sanders, and Peter Smith. 2012. The role of the professional doctorate in developing professional practice. *Journal of Nursing Management* 20(1): 130–139. https://doi.org/10.1111/j.1365-2834.2011.01345.x.

Golde, Chris M. 2005. The role of the department and discipline in doctoral student attrition: Lessons from four departments. *The Journal of Higher Education* 76(6): 669–700. https://doi.org/10.1080/00221546.2005.11772304.

Grathwol, Robert P. 2005. The world through a different window: an international perspective on research. *The Physiologist* 48(4): 169–171.

Hagen, Nils. 2010. Harmonic publication and citation counting: sharing authorship credit equitably–not equally, geometrically or arithmetically. *Scientometrics* 84(3): 785–793. https://doi.org/10.1007/s11192-009-0129-4.

Horta, Hugo, and João M. Santos. 2016. The impact of publishing during PhD studies on career research publication, visibility, and collaborations. *Research in Higher Education* 57:28–50. https://doi.org/10.1007/s11162-015-9380-0.

Horta, Hugo, Sebastian Birolini, Mattia Cattaneo, Wenqin Shen, and Stefano Paleari. 2021. Research network propagation: The impact of PhD students' temporary international mobility. *Quantitative Science Studies* 2(1): 129–154. https://doi.org/10.1162/qss_a_00096.

Hsieh, David. 2013. "Organization and role of international collaboration in research production." M.D. thesis. University of Arizona College of Medicine—Phoenix. http://hdl.handle.net/10150/281178

Huang, Futao. 2010. Transnational higher education in Japan and China: A comparative study. In *Crossing borders in East Asian higher education*, eds. David W. Chapman, William K. Cummings, and Gerard A. Postiglione, 265–282. Dordrecht: Springer Netherlands.

Jazvac-Martek, Marian. 2009. Oscillating role identities: the academic experiences of education doctoral students. *Innovations in Education and Teaching International* 46(3): 253–264. https://doi.org/10.1080/14703290903068862.

Jones, Michael. 2018. Contemporary trends in professional doctorates. *Studies in Higher Education* 43(5): 814–825. https://doi.org/10.1080/03075079.2018.1438095.

Jöns, Heike. 2011. Transnational academic mobility and gender. *Globalisation, Societies and Education* 9(2): 183–209. https://doi.org/10.1080/14767724.2011.577199.

Kehm, Barbara M. 2006. Doctoral education in Europe and North America: A comparative analysis. Wenner Gren International Series 83: 67.

Khalid, Sabrika, Gulnar Orynbek, Cai Lianyu, and Endale Tadesse. 2023. What goes around comes around: Shedding light on today's doctoral student's research socialization and who will be the future faculty. *PLoS ONE* 18(5): e0285843. https://doi.org/10.1371/journal.pone.0285843.

Kiley, Margaret. 2009. "You don't want a smart Alec": Selecting examiners to assess doctoral dissertations. *Studies in higher education* 34(8): 889–903.

Kim, Kiwan, and Steven J. Karau. 2009. Working environment and the research productivity of doctoral students in management. *Journal of Education for Business* 85(2): 101–106. https://doi.org/10.1080/08832320903258535.

Kushkowski, Jeffrey D., Kathy A. Parsons, and William H. Wiese. 2003. Master's and doctoral thesis citations: Analysis and trends of a longitudinal study. *Portal: Libraries and the Academy* 3(3): 459–479. https://doi.org/10.1353/pla.2003.0062.

Kuzhabekova, Aliya, and Jack Lee. 2018. International faculty contribution to local research capacity building: A view from publication data. *Higher Education Policy* 31:423–446. https://doi.org/10.1057/s41307-017-0067-3.

Kwiek, Marek. 2016. The European research elite: a cross-national study of highly productive academics across 11 European systems. *Higher Education* 71(3): 379–397. https://doi.org/10.1007/s10734-015-9910-x.

Kwiek, Marek. 2019. *Changing European academics. A comparative study of social stratification, work patterns and research productivity*. New York: Routledge.

Kwiek, Marek, and Wojciech Roszka. 2021. Gender disparities in international research collaboration: A study of 25,000 university professors. *Journal of Economic Surveys* 35(5): 1344–1380. https://doi.org/10.1111/joes.12395.

Larivière, Vincent. 2012. On the shoulders of students? The contribution of PhD students to the advancement of knowledge. *Scientometrics* 90(2): 463–481.

Larivière, Vincent, Éric Archambault, Yves Gingras, and Étienne Vignola-Gagné. 2006. The place of serials in referencing practices: Comparing natural sciences and engineering with social sciences and humanities. *Journal of the American Society for Information Science and Technology* 57(8): 997–1004.

Lee, Wade M. 2000. Publication trends of doctoral students in three fields from 1965–1995. *Journal of the American Society for Information Science* 51(2): 139–144.

Li, Lechen, and Frank Fernandez. 2024. Mentoring matters: Examining the relationship between adviser interactions and doctoral student publications. *Higher Education*: 1–20. https://doi.org/10.1007/s10734-024-01302-1

Lindahl, Jonas, Cristian Colliander, and Rickard Danell. 2021. The importance of collaboration and supervisor behaviour for gender differences in doctoral student performance and early career development. *Studies in Higher Education* 46(12): 2808–2831. https://doi.org/10.1080/03075079.2020.1861596.

Lubienski, Sarah Theule, Emily K. Miller, and Evthokia Stephanie Saclarides. 2018. Sex differences in doctoral student publication rates. *Educational Researcher* 47(1): 76–81. https://doi.org/10.3102/0013189X17738746.

Mattsson, Pauline, Patrice Laget, Anna Nilsson, and Carl-Johan Sundberg. 2008. Intra-EU vs extra-EU scientific co-publication patterns in EU. *Scientometrics* 75(3): 555–574. https://doi.org/10.1007/s11192-007-1793-x.

McGaskey, Ferlin G. 2015. Facilitating the creation of knowledge: An investigation of the factors that influence the research productivity of Black doctoral students at predominantly White institutions. *Journal of Negro Education* 84(2): 187–201.

McKiernan, Erin C., Lesley A. Schimanski, Carol Muñoz Nieves, Lisa Matthias, Meredith T. Niles, and Juan P. Alperin. 2019. Use of the Journal Impact Factor in academic review, promotion, and tenure evaluations. *eLife* 8:e47338. https://doi.org/10.7554/eLife.47338.

Merton, Robert K. 1973. *The sociology of science: Theoretical and empirical investigations*. Chicago: University of Chicago Press.

Mohaupt, J., M. van Soeren, M.-A. Andrusyszyn, K. MacMillan, S. Devlin-Cop, and S. Reeves. 2012. Understanding interprofessional relationships by the use of contact theory. *Journal of Interprofessional Care* 26(5): 370–375. https://doi.org/10.3109/13561820.2012.673512.

Musselin, Christine. 2004. Toward a European academic labour market? Some lessons drawn from empirical studies on academic mobility. *Higher Education* 48(1): 55–78. https://doi.org/10.1023/B:HIGH.0000033770.24848.41.

Nederhof, Anton J. 2006. Bibliometric monitoring of research performance in the social sciences and the humanities: A review. *Scientometrics* 66(1): 81–100.

Nettles, Michael T., and Catherine M. Millett. 2006. *Three magic letters: Getting to Ph.D*. Baltimore: Johns Hopkins University Press.

Neumann, Ruth. 2005. Doctoral differences: Professional doctorates and PhDs compared. *Journal of Higher Education Policy and Management* 27(2): 173–188. https://doi.org/10.1080/13600800500120027.

Nguyen, Tuan V., Thao P. Ho-Le, and V. Le Ut. 2017. International collaboration in scientific research in Vietnam: an analysis of patterns and impact. *Scientometrics* 110:1035–1051. https://doi.org/10.1007/s11192-016-2201-1.

Nielsen, Mathias Wullum. 2016. Gender inequality and research performance: moving beyond individual-meritocratic explanations of academic advancement. *Studies in Higher Education* 41(11): 2044–2060. https://doi.org/10.1080/03075079.2015.1007945.

Niemczyk, Ewelina Kinga. 2019. Mentorship within doctoral research assistantships: A Canadian case study. *Alberta Journal of Educational Research* 65(3): 221–237.

Onyancha, Omwoyo Bosire. 2021. Regional and international research collaboration and citation impact in selected sub-Saharan African countries in the period 2000 to 2019. *Global Knowledge, Memory and Communication* 70(6/7): 577–594. https://doi.org/10.1108/GKMC-04-2020-0039.

Pettigrew, T. F. 1971. *Racially separate or together?* McGraw-Hill.

Pettigrew, T. F. 1975. The racial integration of the schools. In *Racial Discrimination in the United States*, ed. T. F. Pettigrew, 224–239. Harper & Row.

Pettigrew, T. F. 1998. Intergroup contact theory. *Annual Review of Psychology* 49(1): 65–85. https://doi.org/10.1146/annurev.psych.49.1.65.

Pocock, Barbara, Jude Elton, Deborah Green, Catherine McMahon, and Suzanne Pritchard. 2011. *Juggling work, home, and learning in low paid occupations: A qualitative study.* Adelaide: National Centre for Vocational Education Research. https://www.ncver.edu.au/__data/assets/file/0018/3564/2369.pdf

Pohl, Hans. 2020. Collaboration with countries with rapidly growing research: supporting proactive development of international research collaboration. *Scientometrics* 122(1): 287–307. https://doi.org/10.1007/s11192-019-03287-6.

Powell, Justin J.W., Frank Fernandez, Jon T. Crist, Jennifer Dusdal, Liang Zhang, and David P. Baker. 2017. The worldwide triumph of the research university and globalizing science. In *The century of science: The global triumph of the research university*, eds. Justin J.W. Powell, David P. Baker, and Frank Fernandez, 1–36. Bingley: Emerald.

Ribeiro, Leonardo Costa, Márcia Siqueira Rapini, Leandro Alves Silva, and Eduardo Motta Albuquerque. 2018. Growth patterns of the network if international collaboration in science. *Scientometrics* 114:159–179. https://doi.org/10.1007/s11192-017-2573-x.

Roksa, Josipa, Yapeng Wang, David Feldon, and Matthew Ericson. 2022. Who is publishing journal articles during graduate school? Racial and gender inequalities in biological sciences over time. *Journal of Diversity in Higher Education* 15(1): 47–57. https://doi.org/10.1037/dhe0000336.

Sakamoto, Robin, and David W. Chapman. 2011. *Cross-border partnerships in higher education*. New York: Routledge.

Saliba, Robin. 2012. *PhD program structures in Canada*. Canadian Association of Graduate Studies. https://cags.ca/documents/publications/best_practices/10_12_12_PhD_Program_Structures_in_Canada_Report_R__Saliba.pdf

Savage, William E., and Anthony J. Olejniczak. 2022. More journal articles and fewer books: Publication practices in the social sciences in the 2010's. *PLoS ONE* 17(2): e0263410. https://doi.org/10.1371/journal.pone.0263410.

Schneider, Joanne Kraenzle, Catherine M. Bender, Elizabeth A. Madigan, and Marie T. Nolan. 2020. Facilitating the academic success of international PhD students. *Nursing Education Perspectives* 41(1): 20–25. https://doi.org/10.1097/01.NEP.0000000000000489.

Shin, Jiyun Elizabeth L., Sheri R. Levy, and Bonita London. 2016. Effects of role model exposure on STEM and non-STEM student engagement. *Journal of Applied Social Psychology* 46(7): 410–427. https://doi.org/10.1111/jasp.12371.

Smeby, Jens-Christian, and Jarle Trondal. 2005. Globalisation or europeanisation? International contact among university staff. *Higher Education* 49:449–466. https://doi.org/10.1007/s10734-004-2826-5.

Smith, Robert L., Kenneth Maroney, Kaye W. Nelson, Annette L. Abel, and Holly S. Abel. 2006. Doctoral programs: Changing high rates of attrition. *The Journal of Humanistic Counseling, Education and Development* 45(1): 17–31. https://doi.org/10.1002/j.2161-1939.2006.tb00002.x.

Spector, Paul E. 1994. Using self-report questionnaires in OB research: A comment on the use of a controversial method. *Journal of Organizational Behavior* 15(5): 385–392.

Spence, Nicholas. 2009. Exploring the utility of the 2007 Canadian Graduate and Professional Student Survey student satisfaction at the University of Western Ontario. Ontario: Higher Education Quality Council of Ontario. https://heqco.ca/pub/exploring-the-utility-of-the-2007-canadian-graduate-and-professional-student-survey-student-satisfaction-at-the-university/

Stoilescu, Dorian, and Douglas McDougall. 2010. Starting to publish academic research as a doctoral student. *International Journal of Doctoral Studies* 5:79–92.

The Royal Society. 2011. Knowledge, networks and nations: Global scientific collaboration in the 21st century, RS Policy document 03/11. London: The Royal Society. https://royalsociety.org/-/media/policy/publications/2011/4294976134.pdf

Tregellas, Jason R., Jason Smucny, Donald C. Rojas, and Kristina T. Legget. 2018. Predicting academic career outcomes by predoctoral publication record. *PeerJ* 6:e5707.

Trigwell, Keith, and Harriet Dunbar-Goddet. 2005. *The research experience of postgraduate research students at the University of Oxford*. Oxford: Institute for the Advancement of University Learning, University of Oxford. https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=76aee581d42de015bfcd1c45a5ded78d084e069c

Uddin, Shahadat, Tasadduq Imam, and Mohammad Mozumdar. 2021. Research interdisciplinarity: STEM versus non-STEM. *Scientometrics* 126:603–618. https://doi.org/10.1007/s11192-020-03750-9.

University of Toronto. 2020. Student Guidelines for the Doctoral Thesis. School of Graduate Studies. https://www.sgs.utoronto.ca/current-students/program-completion/producing-your-thesis/doctoral-thesis-guidelines/studentguidelines-for-the-doctoral-thesis/.

Vanstone, Meredith, Kathryn Hibbert, Elizabeth Anne Kinsella, Pamela J. McKenzie, Allan Pitman, and Lorelei Lingard. 2013. Interdisciplinary doctoral research supervision: A scoping review. *Canadian Journal of Higher Education* 43(2): 42–67. https://doi.org/10.47678/cjhe.v43i2.2403.

Wagner, Caroline S., Han Woo Park, and Loet Leydesdorff. 2015. The continuing growth of global cooperation networks in research: A conundrum for national governments. *PLoS ONE* 10(7): e0131816. https://doi.org/10.1371/journal.pone.0131816.

Wagner, Caroline S., Travis A. Whetsell, and Loet Leydesdorff. 2017. Growth of international collaboration in science: Revisiting six specialities. *Scientometrics* 110:1633–1652. https://doi.org/10.1007/s11192-016-2230-9.

Waheed, Syed Abdul. 2022. Navigating the challenges of doctoral journey: Voice of the students pursuing their studies abroad. *Journal of Arts and Social Sciences* 9(2): 100–113. https://doi.org/10.46662/jass.v9i2.254.

Waltman, Ludo. 2012. An empirical analysis of the use of alphabetical authorship in scientific publishing. *Journal of Informetrics* 6(4): 700–711. https://doi.org/10.1016/j.joi.2012.07.008.

Wang, Chi, Jiawei Han, Yuntao Jia, Jie Tang, Duo Zhang, Yintao Yu, and Jingyi Guo. 2010. Mining advisor-advisee relationships from research publication networks. *Proceedings of the 16th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*: 203-212. https://doi.org/10.1145/1835804.18358

Wang, Chuanyi, Fei Guo, and Wu Qing. 2021. The influence of academic advisors on academic network of physics doctoral students: Empirical evidence based on scientometric analysis. *Scientometrics* 126(6): 4899–4925. https://doi.org/10.1007/s11192-021-03974-3.

White, Karen. 2019. Publications output: US trends and international comparisons: Science and engineering indicators 2020(NSB-2020-6). Washington DC: National Science Foundation. https://files.eric.ed.gov/fulltext/ED615534.pdf

Xu, Yonghong Jade. 2013. Career outcomes of STEM and non-STEM college graduates: Persistence in majored-field and influential factors in career choices. *Research in Higher Education* 54:349–382. https://doi.org/10.1007/s11162-012-9275-2.

Yang, Lili, Siqing Li, Yusuf Ikbal Oldac, and Chuanyi Wang. 2023. The agency-structure complex in international research collaboration: Evidence from a Chinese elite university. *Studies in Higher Education* 49:1–14. https://doi.org/10.1080/03075079.2023.2298817.

Yoshioka-Kobayashi, Tohru, and Sotaro Shibayama. 2021. Early career training and development of academic independence: A case of life sciences in Japan. *Studies in Higher Education* 46(12): 2751–2773. https://doi.org/10.1080/03075079.2020.1817889.

Zhao, Huizi. 2012. Exploring the Canadian graduate and professional student survey (CGPSS): Results from 2007 and 2010 for Ontario universities. Ontario: Quality Council of Ontario. https://heqco.ca/wp-content/uploads/2020/03/Final-CGPSS-ENG.pdf

Zippel, Kathrin. 2017. *Women in global science: Advancing academic careers through international collaboration*. Stanford: Stanford University Press.

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